Emotion-Aware Movie Recommendation: An Empirical Study of Benefits and Limitations of Reranking

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Abstract

We present an empirical study of the limitations and diagnostic properties of emotional movie representations for recommendation on MovieLens-1M. The study examines whether interpretable item descriptions translate into useful preference evidence, rather than proposing a new state-of-the-art ranking algorithm. We examine lexical extraction from retrieved synopses, lexical extraction from LLM-generated synopses, and direct LLM scoring of explicit emotional dimensions. The representation analysis motivates the use of direct scoring for downstream experiments while distinguishing descriptive properties from independently validated emotional accuracy. A within-user analysis finds higher mean emotional similarity for relevant candidates for 3,379 of 5,588 examined users, while alignment remains common among other candidates. We then compare mean, cluster-based, and information-gain-based user profiles for weighted reranking of 100 candidates generated by LightGCN and ItemKNN. Relevant items are available beyond the first ten positions for 5,532 of 5,750 LightGCN users, providing an opportunity for refinement. Nevertheless, ranking benefits remain small and depend on the profile and backbone: the information-gain configuration changes LightGCN nDCG@10 from 0.1071 to 0.1073, whereas no profile consistently improves all metrics with ItemKNN. The study shows that explicit emotional representations do not automatically provide an effective preference signal. We discuss lessons concerning the choice of representation, the aggregation of user histories, and the evaluation of incremental ranking utility.

Keywords

recommender systems, emotion-aware recommendation, large language models, reranking, empirical evaluation

1. Introduction

Recommender systems infer preferences from interaction data and item information [1, 2, 3, 4]. For movies, the emotional experience anticipated from a narrative can also influence appeal [5, 6]. Yet an emotional description is not automatically a preference signal: it must be represented, translated from a history into a profile, and add discrimination within a collaboratively retrieved list.

Movie synopses are an accessible source for this analysis. Lexicon-based methods aggregate affective word associations, but depend on lexical coverage and ignore narrative context [7, 8, 9]. LLMs allow context-sensitive, prompt-defined descriptions, although their scores remain inferred characterizations rather than measurements of viewers' responses. We build on Hill et al. [10], adapting their emotion-vector perspective from book descriptions and age groups to movie synopses and individual histories.

The two transitions are methodologically distinct. Item-level extraction asks whether a representation is inspectable, sufficiently covered, and meaningful for the collection. Profiling asks whether the same representation summarizes a history whose movies may have been appreciated for different reasons and on different occasions. A useful representation may thus fail to distinguish relevant candidates, especially after collaborative retrieval has already concentrated plausible items near the top of the list. Treating these transitions separately is central to the empirical question addressed here.

This paper is a diagnostic study, not a new state-of-the-art ranking method. We compare lexical extraction from retrieved and LLM-generated synopses with direct LLM scoring, then rerank candidates

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from LightGCN [11, 12] and ItemKNN [13] using mean, cluster-based, and information-gain-based profiles.

We address three questions:

- **RQ1:** How do the representations differ in descriptive properties?
- **RQ2:** Do they reveal preference-related patterns in user histories?
- **RQ3:** How does profile construction affect reranking utility?

The contribution connects representation, preference evidence, and ranking utility. Direct scoring is selected for operational properties, not established emotional accuracy. The observed ranking changes are marginal and configuration-dependent, showing where emotional enrichment supplies interpretable structure and where it remains insufficient for reliable refinement.

2. Related Work

Emotion-aware recommendation can model item characteristics, user states, or elicited responses [10, 14]; we study only synopsis-derived item descriptions and infer preferences from rating histories. Hill et al. [10] aggregate NRC-EIL scores from book descriptions, expand coverage with WordNet synonyms, and use emotion vectors in LLM fine-tuning. We adapt that description-based perspective to movies and compare it with direct LLM scoring, whose task-specific vocabulary and contextual inference come with dependence on model, prompt, and input [15, 16, 17].

Movie-focused affective recommendation has also represented films in connotative or affective feature spaces. Benini et al. [18] define a connotative space for movie retrieval and recommendation, while Canini et al. [19] use selected connotative features for affective movie recommendation. Those approaches motivate the use of affect as an item-side signal, but they do not answer whether a synopsis-derived emotional description can refine a modern collaborative candidate list. Our study addresses this narrower multistage question and reports both the available headroom and the limited observed gains.

Recent systems use LLMs as rankers over retrieved candidates [20, 21, 22] or jointly learn textual and collaborative representations [23, 24, 25]. Our setting is narrower: the LLM supplies one explicit affective facet, while a collaborative model retrieves candidates. We therefore test whether that facet adds discrimination in a fixed candidate pool, rather than propose a new retrieval or LLM-ranking architecture.

This distinction also separates our work from recommendation approaches based on explicit affective goals. For example, Hasan and Bunescu [14] learn from fine-grained affective states expressed in reviews. Here, no user states or affective requests are observed: emotional preference is inferred only from interactions. The weaker supervision makes the setting broadly applicable to rating data, but it also makes a positive ranking effect more difficult to interpret as evidence of an intended emotional experience.

3. Methodology

We construct emotional item vectors, aggregate them into user profiles, and combine emotional compatibility with collaborative scores to rerank retrieved candidates.

3.1. Problem formulation and architecture

Let $\mathcal{U}$ and $\mathcal{I}$ denote the users and items, and let $H_u \subseteq \mathcal{I}$ be the observed history available for user u. A collaborative recommender assigns a user–item relevance score $s(u, i)$ and retrieves the k highest-scoring unseen items, forming a candidate set $C_u \subseteq \mathcal{I} \setminus H_u$. Each item has an emotional vector $\mathbf{e}_i$, and a profiling function constructs a user representation P_u from the vectors of items in H_u. An emotional

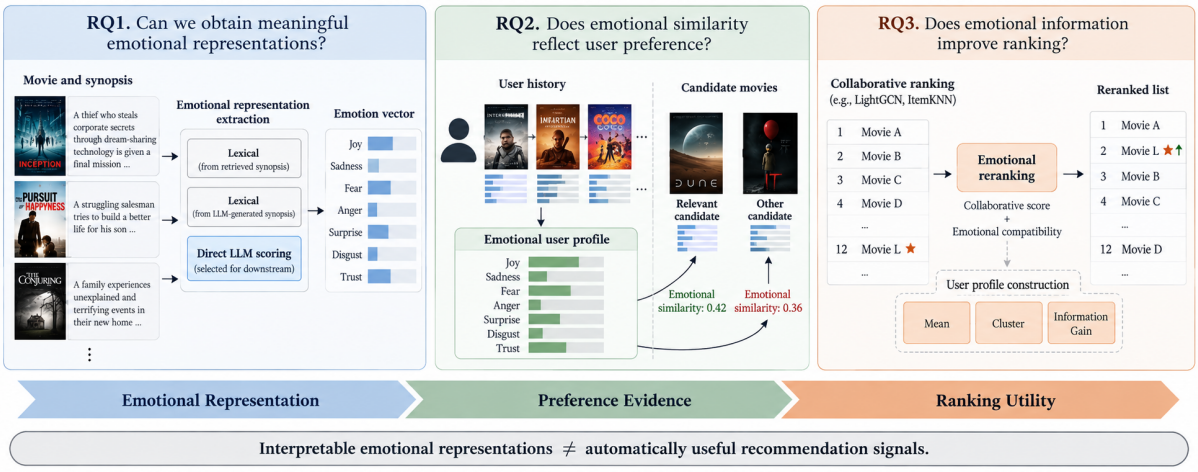


Figure 1: Study overview: representations (RQ1), preference evidence (RQ2), and reranking utility (RQ3). Examples are illustrative.

Table 1
Representation strategies. The lexical variants use eight NRC categories, optionally augmented with an unmatched-token component. Direct scoring uses seven explicitly defined categories.

Strategy	Textual input	Extraction	Output
Static	Retrieved synopsis	NRC lexical matching	Normalized lexical weights
Hybrid	LLM-generated synopsis	Same lexical procedure	Normalized lexical weights
Direct LLM	Generated synopsis with movie metadata	Explicit emotional scoring	Seven scores from 1 to 5

compatibility function g compares this representation with a candidate vector, giving $c(u,i) = g(P_u, \mathbf{e}_i)$. The final score is defined generically as

$$r(u,i) = F\big(s(u,i), g(P_u, \mathbf{e}_i)\big), \qquad i \in C_u. \tag{1}$$

Here, P_u includes any selected emotional coordinates, and g includes the preprocessing needed for comparison. The following subsections instantiate the item representations and user profiles; Section 3.4 specifies F as a weighted combination of normalized scores.

Figure 1 summarizes the pipeline. The reranker changes only the order of items already in C_u.

3.2. Emotional item representations

Table 1 outlines the strategies. They are design alternatives, not a single-factor comparison: direct scoring changes taxonomy, scale, input and extraction method.

Static lexical extraction. Following Hill et al. [10], retrieved synopses are normalized, tokenized, stripped of stopwords, and matched to NRC-EIL with WordNet synonym expansion [26, 27, 28]. Scores for **anger**, **anticipation**, **disgust**, **fear**, **joy**, **sadness**, **surprise**, and **trust** are summed and normalized. *Expressionless* records unmatched evidence, analogous to the Objective component in the source approach, not emotional neutrality.

Hybrid lexical extraction. Hybrid extraction applies the same procedure to an LLM-generated neutral synopsis. It may increase lexical evidence, but also changes the content and wording; its effects cannot be attributed to length alone.

Direct LLM scoring. Direct scoring uses the generated synopsis and metadata to score **joy**, **sadness**, **anger**, **suspense**, **fear**, **wonder**, and **disgust** on a 1–5 scale in JSON. It retains five NRC labels but replaces anticipation, surprise and trust with suspense and wonder; it is therefore not a reproduction of the lexical taxonomy.

The prompt defines each coordinate in narrative terms and asks the model to use only the supplied title, year, genres, and plot. In particular, suspense refers to tension and uncertainty, fear to threat or menace, and wonder to awe, discovery, or the extraordinary. The prompt does not require the seven scores to sum to a fixed value: a film can therefore be both suspenseful and fearful, or show several weak dimensions. This choice preserves potentially co-occurring narrative tones, but it does not turn the generated scores into human annotations.

We use direct scoring downstream for its explicit, adaptable dimensions, not established emotional accuracy. Missing outputs are imputed with dimension-wise means; this preserves interactions but may attenuate emotional variation. The representation is standardized before profile construction, so its downstream use depends on relative variation within each dimension rather than the absolute interpretation of a raw score alone.

The three representations therefore answer different practical needs. Static lexical extraction is transparent but vulnerable to sparse synopses; hybrid extraction tests whether generated narrative material improves lexical coverage; direct scoring makes the intended dimensions explicit to the model. The subsequent experiments do not identify a universally best extractor. They examine whether the direct representation, selected as a flexible operational input, can be translated into a ranking signal.

3.3. From item vectors to user profiles

Let $\mathbf{z}_i$ denote the standardized emotional vector:

$$z_{ij} = \frac{e_{ij} - \mu_j}{\sigma_j}, \tag{2}$$

where μ_j and σ_j are the dimension's reference mean and standard deviation. The selection-based strategy additionally uses a binary representation for estimating information gain; cosine similarity itself is computed on the standardized continuous scores.

Mean profile. The full-history profile is

$$\mathbf{p}_u = \frac{1}{|H_u|} \sum_{i \in H_u} \mathbf{z}_i. \tag{3}$$

Similarity is

$$c(u,i) = \frac{\mathbf{p}_u^\top \mathbf{z}_i}{\|\mathbf{p}_u\|_2 \|\mathbf{z}_i\|_2}. \tag{4}$$

The mean retains every observed item with equal weight. It is a deliberately simple reference profile: it can reveal broad tendencies, but may smooth distinct emotional regions of an interaction history. The cluster and information-gain variants modify this assumption in complementary ways, by selecting items or dimensions respectively.

Cluster-based profile. HDBSCAN is a hierarchical density-based method building on DBSCAN [29]. We use Euclidean distance and minimum cluster size $\max(3, \lceil 0.05|H_u| \rceil)$; other parameters use defaults. The non-noise cluster with greatest mean membership strength is represented by its centroid. If none exists, the collaborative order is retained.

Information-gain selection. Observed and equally sized sampled-unobserved item sets define Y_u. For binary presence B_j, direct scores > 2.5 are one; other scores are zero. Ten samples (seed 42) are averaged. The information gain is

$$\mathrm{IG}_u(j) = H(Y_u) - H(Y_u \mid B_j), \tag{5}$$

Emotions are retained in decreasing gain order until cumulative gain exceeds 0.3, producing S_u; otherwise S_u is empty. Compatibility uses the selected coordinates:

$$c_{S_u}(u,i) = \frac{\mathbf{p}_{u,S_u}^{\top}\mathbf{z}_{i,S_u}}{\|\mathbf{p}_{u,S_u}\|_2\|\mathbf{z}_{i,S_u}\|_2}. \tag{6}$$

An empty selection preserves the collaborative order. Unobserved items are not confirmed dislikes, and we do not test sensitivity to the selection settings.

The selected subspace is user-specific, but the profile itself remains the mean over historical standardized vectors. Information gain consequently changes the coordinates used for comparison rather than creating a second latent user model. This distinction matters when interpreting its behavior: a small aggregate change can result from limited selected coordinates, the fallback, or the conservative score fusion.

3.4. Weighted reranking

To combine scores with different ranges, we normalize the collaborative score $s(u,i)$ and emotional compatibility $c(u,i)$ separately within C_u:

$$\tilde{s}(u,i) = \frac{s(u,i) - \min_{j\in C_u} s(u,j)}{\max_{j\in C_u} s(u,j) - \min_{j\in C_u} s(u,j)}, \tag{7}$$

$$\tilde{c}(u,i) = \frac{c(u,i) - \min_{j\in C_u} c(u,j)}{\max_{j\in C_u} c(u,j) - \min_{j\in C_u} c(u,j)}. \tag{8}$$

Constant score ranges and zero cosine denominators are mapped to zero; ties use normalized collaborative score. The final score is

$$r(u,i) = \alpha\,\tilde{s}(u,i) + (1-\alpha)\,\tilde{c}(u,i). \tag{9}$$

The parameter $\alpha \in [0,1]$ controls the collaborative contribution; normalization is local to each candidate list.

4. Experimental Setting

4.1. Dataset and evaluation protocol

We use MovieLens-1M [30], a well-established benchmark in recommender systems research [31, 32]. We retain ratings ≥ 3 and apply iterative 20-user/10-item filtering, obtaining 829,683 interactions, 5,750 users, and 3,122 items. We use DataRec and Elliot [33, 34]. Timestamp splits are 70/10/20 for training/validation/test; profiles use pre-test histories. This is a rating-history evaluation, not a reconstruction of emotional states at viewing time.

4.2. Text sources and emotion extraction

Static synopses come from OMDb [1]. Hybrid and direct descriptions use Gemini 2.5 Flash at temperature 0, 500 output tokens, and disabled thinking. The generation prompt requests a neutral 120–200 word plot and permits `None`; the scoring prompt supplies title, year and plot, defines the seven dimensions, and requires JSON scores from 1 to 5. Templates and parsing are in the accompanying code. No seed or repeated-call agreement study is reported.

[1] https://www.omdbapi.com/

4.3. Candidate generators and profile configurations

LightGCN [11] and ItemKNN [13] generate $k = 100$ candidates. LightGCN tunes learning rate, regularization and depth by validation nDCG@10; ItemKNN uses cosine similarity and tunes neighborhood size.

We compare mean, cluster and information-gain profiles using direct LLM representations. The reported fusion uses $\alpha = 0.9$ and IG threshold 0.3, a conservative exploratory setting that gives 90% weight to collaborative score. It is not validation-optimized and we do not claim that other weights are ineffective.

The multistage protocol intentionally fixes the candidate set before applying emotional information. This isolates whether compatibility can refine an existing collaborative order, but cannot test whether emotional features would improve retrieval itself. The setting also means that a reranker can only promote a relevant item when the backbone has already retrieved it.

4.4. Analysis of emotional patterns

For RQ2, let T_u be held-out relevant items, $R_u = C_u \cap T_u$, and $O_u = C_u \setminus T_u$. We compare mean-profile similarity for users with both groups. Items in O_u are unobserved candidates, not confirmed dislikes.

For each user, we inspect the differences

$$\Delta_u^{\text{mean}} = \frac{1}{|R_u|} \sum_{i \in R_u} c(u,i) - \frac{1}{|O_u|} \sum_{i \in O_u} c(u,i), \tag{10}$$

$$\Delta_u^{\max} = \max_{i \in R_u} c(u,i) - \max_{i \in O_u} c(u,i). \tag{11}$$

We report distributions and positive differences; labels are used only for this descriptive diagnostic.

We also inspect demographic composition for favorable alignment using

$$\Delta_u^{\min} = \min_{i \in R_u} c(u,i) - \min_{i \in O_u} c(u,i). \tag{12}$$

Metadata never enter profiling or reranking; this is exploratory, and extrema depend on group size.

The metadata analysis is retained as a diagnostic of heterogeneity, not as a user model. In particular, the groups are selected using held-out relevance and cannot provide a valid basis for demographic routing or causal interpretation.

4.5. Evaluation measures

We report binary-relevance nDCG@1, @10 and @100, plus MRR@100. Results are exploratory point estimates: no significance tests, confidence intervals, or candidate-pool sensitivity analyses establish superiority or reliability.

5. Results

5.1. RQ1: from lexical evidence to explicit emotional scores

OMDb retrieval succeeds for 3,425 of 3,883 catalogue entries, with 22-word synopses on average. Generated synopses reduce missing lexical evidence: for disgust, zero components fall from about 45% to 2.17%. They do not produce clearer leading lexical components, however: the mean top-two gap is 0.1044 for static and 0.0577 for hybrid representations.

The static representation is additionally dominated by unmatched evidence: its *Expressionless* component has mean 0.5546 and is the largest component for 98.57% of films. Removing that component makes the lexical weights easier to inspect, but does not create sharply distinctive profiles. Table 2 shows that the mean weights of static and hybrid extraction remain close. The hybrid approach therefore solves a coverage problem rather than demonstrating a more discriminative emotional representation.

Table 2
Mean lexical weights after excluding the unmatched-evidence component. Hybrid text improves coverage but preserves a similarly diffuse distribution across NRC dimensions.

	Ang.	Ant.	Dis.	Fear	Joy	Sad.	Sur.	Trust
Static	.104	.135	.060	.166	.165	.109	.051	.210
Hybrid	.113	.148	.058	.160	.140	.101	.057	.223

Table 3
Within-user differences in profile–candidate cosine similarity: the statistic for relevant candidates minus the same statistic for other candidates. Quartiles refer to 5,588 users with at least one relevant top-100 candidate.

Statistic	First quartile	Median	Third quartile	Users with difference > 0
Mean	−0.1062	0.0653	0.2355	3,379
Maximum	−0.2779	−0.1285	−0.0477	452
Minimum	0.1747	0.4431	0.8846	5,206

Direct scoring provides explicit, domain-oriented dimensions. Its scores show interpretable structure (suspense–fear correlation 0.78; joy–fear −0.63), motivating its downstream use. These properties do not establish label accuracy: the alternatives differ in categories, scale, input and extraction, without independent ground truth.

Other observed relations support the same limited interpretation. Joy is negatively associated with sadness (−0.50), anger (−0.57), and disgust (−0.64), whereas suspense is moderately associated with anger and fear (both 0.53). These patterns indicate that the prompt produces related but non-identical coordinates instead of a uniform score vector. They are internal properties of the generated collection: they neither validate the emotional meaning of individual films nor show that viewers experience the corresponding emotions.

The representation findings thus answer RQ1 at the level needed for the pipeline. Generated text makes lexical matching less sparse, while direct scoring produces a compact space whose dimensions have recognizable aggregate relations. Neither observation is a controlled accuracy comparison. In particular, a correlation between two scores shows a structured representation, not agreement with annotations of viewers or experts.

5.2. RQ2: emotional patterns in user histories

We use direct scores and the mean profile for 5,588 LightGCN users with at least one relevant top-100 candidate (42,838 relevant and 515,962 other pairs). Positive cosine similarity occurs for 71.3% and 68.9% of these groups, respectively, so sign alone offers little separation. Relevant candidates have higher mean similarity for 3,379 users; Table 3 shows the heterogeneous distribution.

Only 452 users show a positive maximum difference; the approximately twelve-fold prevalence of other pairs makes this extreme statistic strongly group-size dependent. The positive minimum difference for 5,206 users is easier to obtain: with many more other candidates, there are more opportunities to observe one with very low compatibility. It should therefore be read as evidence that some relevant candidates avoid the worst emotional matches, rather than as a useful promotion rule. Thus, RQ2 identifies a limited, heterogeneous association rather than an emotion-only separator or a population-wide psychological pattern.

The within-user framing is important. It compares relevant and other candidates for the same profile instead of treating all candidates as exchangeable across users. Even so, the relevant group is much smaller, the items in O_u are not dislikes, and candidates have already been filtered by LightGCN. The observed association is therefore evidence about the signal available after retrieval, not a standalone evaluation of emotional preference prediction.

Metadata subsets with favorable alignment contain relatively more users in categories 18 and 25,

Table 4
Location of held-out relevant items in LightGCN candidate lists. Counts are user–item occurrences, not distinct movies across users.

Positions	Relevant user–item pairs	Users with at least one relevant item
1–100	42,838	5,588
1–10	6,686	3,262
11–100	36,152	5,532

Table 5
Ranking performance before and after emotional reranking. All rerankers operate on the candidate pool of the corresponding backbone. Values are point estimates; no statistical significance is claimed.

Backbone	Profile / configuration	nDCG@1	nDCG@10	nDCG@100	MRR@100
LightGCN	Collaborative baseline	0.1065	0.1071	0.2135	0.2774
	Random reranking	0.0453	0.0561	0.1832	0.1946
	Mean profile	0.0985	0.1066	0.2127	0.2734
	Cluster-based profile	0.1013	0.1057	0.2124	0.2723
	Information-gain selection	0.1066	0.1073	0.2137	0.2776
ItemKNN	Collaborative baseline	0.1186	0.1169	0.2125	0.2911
	Random reranking	0.0453	0.0561	0.1832	0.1946
	Mean profile	0.1175	0.1171	0.2127	0.2906
	Cluster-based profile	0.1192	0.1169	0.2126	0.2915
	Information-gain selection	0.1185	0.1170	0.2126	0.2910

artists and college/graduate students, and fewer executive/managerial users; gender shifts are small. These are exploratory composition patterns, not causal effects or demographic targeting rules.

5.3. RQ3: recommendation utility of emotional profiles

Candidate availability. For LightGCN, 36,152 of 42,838 relevant pairs occur at positions 11–100, covering 5,532 users (Table 4). Thus relevant candidates exist below rank ten, although this does not establish attainable reranking gains.

Table 5 reports point estimates for each backbone and profile.

For LightGCN, mean and cluster profiles reduce every metric; IG produces only small favorable changes, including nDCG@10 from 0.1071 to 0.1073. For ItemKNN, no profile improves every metric: mean has the highest nDCG@10 and clustering the highest nDCG@1 and MRR@100. RQ3 therefore finds no uniformly effective strategy; relevant candidates below rank ten do not make emotional compatibility sufficient for promotion.

Random reranking provides a descriptive lower bound for this comparison. Its substantial degradation confirms that the initial collaborative orders contain useful positional information and that a perturbation is not neutral. The emotional configurations remain much closer to their respective backbones than to this random condition, but this does not establish a practically material benefit: most changes are still small relative to the baseline metrics and are not accompanied by uncertainty estimates.

The results also show why a candidate-availability check is necessary but insufficient. Many users have relevant items outside the top ten, leaving scope for an effective reranker to act. Yet the profiles fail to identify those candidates consistently enough to yield material aggregate improvements. The contrast between LightGCN and ItemKNN further indicates that a profile cannot be evaluated independently of the collaborative order it refines.

6. Discussion and Lessons Learned

6.1. Lessons learned

Richer synopses alleviate missing lexical evidence, and direct scoring gives flexible, interpretable dimensions; neither property guarantees ranking value. RQ2 finds higher mean alignment for relevant candidates for some users, but positive similarity is common in both groups and the association is heterogeneous. The observed demographic composition patterns are exploratory and cannot justify demographic targeting.

This separation between description and utility is the main lesson of the study. A movie representation can be useful for inspection, explanation, collection analysis, or filtering without being sufficiently discriminative for top-k ranking. Conversely, a small ranking change cannot validate emotional fidelity. Representation quality and recommendation utility therefore require different evidence: annotation or agreement studies for the former, and controlled ranking comparisons for the latter.

Profile construction is consequential. A mean can cancel contrasting experiences, clustering discards history, and IG masks dimensions; the fallback also preserves the baseline for some users. Yet the results do not isolate these mechanisms. Relevant candidates below rank ten show that limited gains are not explained only by missing candidates. A more likely difficulty is that interaction data already encode part of the information in synopsis-derived affect, while inference and aggregation can add noise. The conservative collaborative weight further limits the intervention.

The profile variants should consequently be understood as analytical probes. The mean asks whether an overall emotional tendency is useful; clustering tests whether one coherent historical subset is more representative; information gain tests whether only selected dimensions should contribute. Their different results across backbones show that there is no single natural translation from emotional item vectors to user preference. The fact that IG stays close to the baseline is compatible with selective dimensions and fallback, but the present study does not quantify the contribution of either mechanism separately.

6.2. Limitations

The study uses one dataset, a top-100 pool, point estimates, and alternatives that change taxonomy, scale, input and extraction together. Emotional labels have no independent ground truth; ratings do not measure emotional states; and unobserved candidates are not dislikes. We do not test significance, uncertainty, candidate-pool sensitivity, semantic or genre baselines, valence–arousal features, or learned rerankers. Future work should add these controls, shared-input representation tests, independent annotation, and multiple datasets before claiming incremental value beyond generic content features.

A particularly useful next evaluation would compare emotional compatibility with controls that preserve the collaborative component: shuffled item–emotion assignments, generic synopsis embeddings, and genre similarity. User-level bootstrap intervals could distinguish numerical noise from small but reliable effects. A learned reranker may also exploit interactions between collaborative and affective evidence more effectively than fixed convex fusion, but it should be evaluated against these controls rather than assumed to solve the representation problem.

7. Conclusion

We investigated the use of explicit emotional representations in multistage movie recommendation. The representation analysis motivates direct LLM scoring as a flexible, interpretable input to profiling, while distinguishing this operational choice from a demonstration of superior emotional accuracy. The analysis of user histories identifies heterogeneous associations between emotional profile similarity and held-out candidate relevance. Exploratory metadata associations further characterize this heterogeneity, without establishing demographic causes or a validated targeting rule. The downstream comparison

then examines how full-history averaging, cluster selection, and information-gain selection influence reranking with two collaborative backbones.

The results show small, configuration-dependent benefits despite the presence of relevant items below the initial top ten. No profiling strategy consistently improves every metric across both backbones. The central lesson is that turning emotional item descriptions into useful preference evidence requires its own evaluation: representation, aggregation, and the collaborative ordering all affect the outcome. These findings support a measured role for emotional reranking and identify where stronger evidence is needed to establish its incremental value.

Declaration on Generative AI

Gemini 2.5 Flash was used in the experimental pipeline to generate movie synopses and assign direct intensity scores to the seven emotional dimensions defined in the prompt. Generated synopses also served as input to the hybrid lexical extraction procedure. These outputs constitute inferred item representations rather than independently validated emotional annotations. OpenAI Codex was used for manuscript language revision and LaTeX editing. The authors reviewed and edited the AI-assisted content as appropriate and take full responsibility for the scientific content, references, interpretation of the experimental evidence, and final manuscript.

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